CO-1 AT-1

Name : V. Venkata Susheel Chowdary

Reg no. : 192472060

Course code : ITA0502

Course name : Computer Vision

1. The problem is low spacial sampling, meaning the image does not contain enough pixels to represent small objects clearly.

2. We use histogram equalization to enhance the brightn-ess and contrast of the image. A noise-reduction filter can reduce the noise.

3. Quantization converts continuous intensity values into finite no. of digital intensity levels. An image can use 256 levels from 0 to 255. This allows continuous image to be processed digitally.

4. We use contrast enhancement such as histogram equalization. It improves the contrast between the product and its defects making defects easier to identify even when lighting is uneven.

5. Image segmentation divides the image into meaningful regions such as roads, vehicles and pedistrains. This makes self-driving system identify objects and make decisions.

6. We can use Canny edge detector to identify sharp edges, intensity changes corresponding to boundaries and cracks.

7. A median filter can be applied to remove the noise while preserving important edges. This produces a cleaner medical image, making the tumor and it's boundaries easier to observe.

8. We can apply the object detection algorithm to locate vehicles and draw boundary boxes around them.

9. The blur mainly caused by object movement during image acquistion. Image restoration techniques can be applied to reduce the blur and recover a sharper image.

10. Train an image classification model using images of good and defective products. The model classifies each new product as defective or good.